

August 9, 2026

Dear Hiring Team,

I am writing to apply for the Junior Data Scientist position with mthree. With a Master of Science in Computer Science from the University of Calgary and hands-on experience building Python/C++ machine learning systems, analytical data pipelines, NLP-related backend services, and model evaluation workflows, I am excited by the opportunity to support data-driven trading, risk, and regulatory analytics.

My graduate research focused on turning large, complex datasets into structured, queryable, and reliable analytical systems. Working with BigGeo, Mitacs, and the University of Calgary, I built Python and C++ software for ingesting, preprocessing, storing, retrieving, and evaluating Earth observation data. I developed structured tensor-based data models, automated preprocessing pipelines, documented modeling assumptions, and improved one C++ processing workflow from 233 seconds to 26 seconds for a 10-tensor workload.

That research background aligns well with mthree's need for someone comfortable with Python, pandas, NumPy, machine learning algorithms, predictive analysis, and careful analytical reasoning. In my DEM super-resolution project, I trained and evaluated PyTorch models, processed large geospatial datasets, and used quantitative metrics such as RMSE, slope, terrain ruggedness, topographic position, and wavelet-based errors to understand model performance. This strengthened my ability to move from raw data to defensible modeling results, not just code that runs.

I also bring practical NLP and backend experience. As a back-end developer at Asr Gooyesh Pardaz, I built Django REST APIs for NLP, text-to-speech, and speech-to-text service plans backed by PostgreSQL, implemented authentication flows, and contributed to a chatbot system using GPT-style responses and FAQ data. That work gave me a useful bridge between language-processing systems, relational data, APIs, and user-facing software.

Although I have not worked directly in fixed-income or FX trading analytics, I am genuinely interested in learning the domain and applying my quantitative and software background to financial data problems. I am comfortable ramping up quickly, asking precise questions, documenting assumptions, and communicating technical results clearly. My teaching assistant experience in big data applications, data-at-scale workflows, SQL, and deep learning has also trained me to explain complex ideas and support others through technical ambiguity.

Thank you for considering my application. I would welcome the opportunity to discuss how my Python, machine learning, NLP, SQL, data-pipeline, and analytical modeling experience can contribute to mthree's client and trading analytics work.

Sincerely,
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